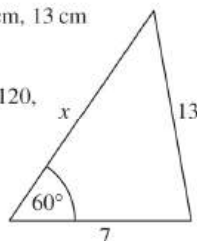


2001

Q1

The diagram shows a triangle with sides 7 cm, 13 cm and x cm, and an angle of 60° as marked.

Use the cosine rule to show that $x^2 - 7x = 120$, and hence find the exact value of x .

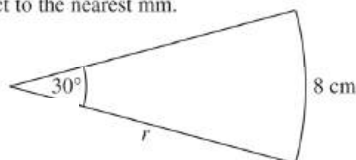


$$\begin{aligned} a^2 &= b^2 + c^2 - 2bc \cos A \\ 13^2 &= 7^2 + x^2 - 2 \times 7 \times x \times \cos 60^\circ \\ 169 &= 49 + x^2 - 7x, \quad \left(\cos 60^\circ = \frac{1}{2} \right) \\ \therefore x^2 - 7x &= 120. \\ x^2 - 7x - 120 &= 0 \\ (x - 15)(x + 8) &= 0 \\ \therefore x &= 15 \text{ or } x = -8. \\ \text{Since } x &\text{ is a length, } x > 0. \\ \therefore x &= 15. \end{aligned}$$

2001

Q2

Find the length of the radius of the sector of the circle shown in the diagram. Give your answer correct to the nearest mm.

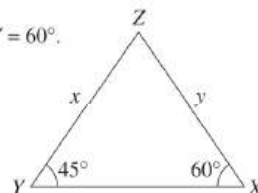


$$\begin{aligned} \ell &= r\theta; \quad \ell = 8, \quad \theta = \frac{\pi}{6} \\ 8 &= r \times \frac{\pi}{6} \\ r &= \frac{48}{\pi} \\ &= 15.278 \dots \\ &= 15.3 \text{ cm (153 mm)}. \end{aligned}$$

2002

Q3

In the diagram, XYZ is a triangle where $\angle ZYX = 45^\circ$ and $\angle ZXY = 60^\circ$.



$$\begin{aligned} \text{Find the exact value for the ratio } \frac{x}{y}. \\ \frac{x}{\sin 60^\circ} &= \frac{y}{\sin 45^\circ} \\ \therefore \frac{x}{y} &= \frac{\sin 60^\circ}{\sin 45^\circ} \\ &= \frac{\frac{\sqrt{3}}{2}}{\frac{\sqrt{2}}{2}} \\ &= \frac{\sqrt{3}}{\sqrt{2}} \\ &= \frac{\sqrt{6}}{2}. \end{aligned}$$

4

2002

Q4

In the diagram, LMN is a triangle where $LM = 5.2$ metres, $LN = 8.9$ metres and angle $MLN = 110^\circ$.

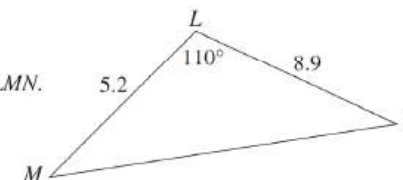
(i) Find the length of MN .

(ii) Calculate the area of triangle LMN .

(i) By the cosine rule:

$$\begin{aligned} MN^2 &= LM^2 + LN^2 - 2(LM)(LN) \cos \angle MLN \\ &= 5.2^2 + 8.9^2 - 2(5.2)(8.9) \cos 110^\circ \\ &= 137.9073 \dots \quad (\text{by calc.}) \\ \therefore MN &= 11.7433 \dots \quad (\text{by calc.}) \\ &\approx 11.7 \text{ m} \end{aligned}$$

$$\begin{aligned} \text{(ii) Area } \triangle LMN &= \frac{1}{2} (LM)(LN) \sin \angle MLN \\ &= \frac{1}{2} \times 5.2 \times 8.9 \times \sin 110^\circ \\ &= 21.7444 \dots \quad (\text{by calc.}) \\ &\approx 21.7 \text{ m}^2. \end{aligned}$$



2002

Q5

Find all values of θ , where $0^\circ \leq \theta \leq 360^\circ$, that satisfy the equation $\cos \theta - \frac{2}{5} = 0$. 2
Give your answer(s) to the nearest degree.

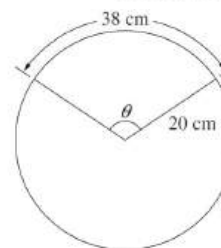
$$\begin{aligned} \cos \theta - \frac{2}{5} &= 0 \quad 0^\circ \leq \theta \leq 360^\circ \\ \therefore \cos \theta &= \frac{2}{5} \\ \therefore \theta &= 66.421 \dots^\circ \text{ or } 360^\circ - 66.421 \dots \\ \therefore \theta &= 66^\circ \text{ or } 293.578 \dots^\circ \\ &= 66^\circ \text{ or } 294^\circ \text{ to nearest degree.} \end{aligned}$$

2002

Q6

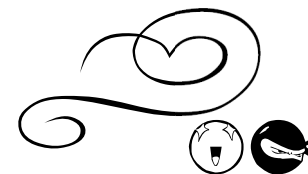
The length of the arc between two spokes on a car's steering wheel is 38 cm. Each spoke is 20 cm in length. 2

Calculate the angle θ between the two spokes. Give your answer correct to the nearest degree.



$$\begin{aligned} l &= r\theta \\ \therefore 38 &= 20 \times \theta \\ \theta &= \frac{38}{20} \\ &= 1.9 \text{ radians} \\ \text{Now } 1.9 \text{ radians} &= \left(1.9 \times \frac{180}{\pi} \right)^\circ \\ &= 108.8619 \dots^\circ \quad (\text{by calc.}) \\ &= 109^\circ \text{ to nearest degree.} \end{aligned}$$

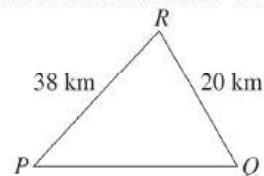




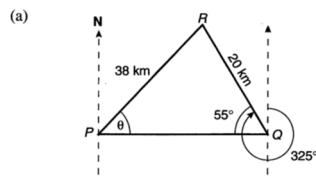
2003

Q7

In the diagram, the point Q is due east of P . The point R is 38 km from P and 20 km from Q . The bearing of R from Q is 325° .



- What is the size of $\angle PQR$?
- What is the bearing of R from P ?



(i) $\angle PQR = 325^\circ - 270^\circ = 55^\circ$.

(ii) By the sine rule in $\triangle PQR$:

$$\frac{\sin \theta}{20} = \frac{\sin 55^\circ}{38}$$

$$\begin{aligned}\sin \theta &= \frac{20 \times \sin 55^\circ}{38} \\ &= 0.4311 \dots \quad (\text{by calc.}) \\ \therefore \theta &= 25.539 \dots^\circ \quad (\text{by calc.}) \\ &= 25^\circ 32'\end{aligned}$$

$$\begin{aligned}\therefore \angle NPR &= 90^\circ - 25^\circ 32' \\ &= 64^\circ 28'\end{aligned}$$

$$\begin{aligned}\therefore \text{Bearing of } R \text{ from } P &= \angle NPR \\ &= 64^\circ.\end{aligned}$$

2003

Q8

Solve $2 \sin^2 x - 3 \sin x - 2 = 0$ for $0 \leq x \leq 2\pi$.

$$\begin{aligned}2 \sin^2 x - 3 \sin x - 2 &= 0 \quad 0 \leq x \leq 2\pi \\ (2 \sin x + 1)(\sin x - 2) &= 0 \\ \therefore 2 \sin x + 1 &= 0 \text{ or } \sin x - 2 = 0\end{aligned}$$

$$\therefore \sin x = -\frac{1}{2} \text{ or } \sin x = 2$$

$\sin x = 2$ has no real solutions
since $-1 \leq \sin x \leq 1$.

$$\therefore \sin x = -\frac{1}{2}$$

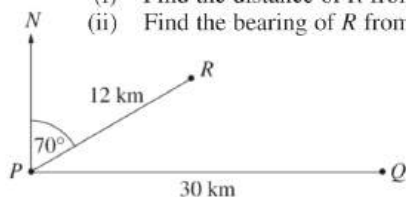
$$x = \left(\pi + \frac{\pi}{6}\right) \text{ or } \left(2\pi - \frac{\pi}{6}\right)$$

$$= \frac{7\pi}{6} \text{ or } \frac{11\pi}{6}.$$

2004

Q9

The diagram shows a point P which is 30 km due west of the point Q . The point R is 12 km from P and has a bearing from P of 070° .



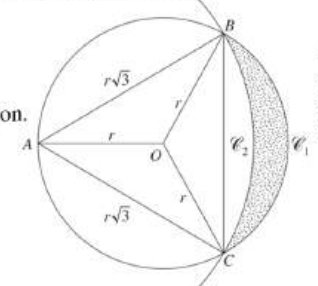
- Find the distance of R from Q .
- Find the bearing of R from Q .

2003

Q10

The point A lies on the circle $\mathcal{C}_1$ with centre O and radius r . The circle $\mathcal{C}_2$ with centre A and radius $r\sqrt{3}$ intersects $\mathcal{C}_1$ at the points B and C , as shown in the diagram.

- Using the cosine rule (or otherwise) find $\angle BAO$.
- Find the area of the sector BAC of $\mathcal{C}_2$.
- Find the area of the sector BOC of $\mathcal{C}_1$.
- Hence (or otherwise) find the area of the shaded region.



$$\begin{aligned}\text{(i)} \quad \cos \angle BAO &= \frac{r^2 + (r\sqrt{3})^2 - r^2}{2(r)(r\sqrt{3})} \\ &= \frac{r^2 + 3r^2 - r^2}{2\sqrt{3}r^2} \\ &= \frac{3r^2}{2\sqrt{3}r^2} \\ &= \frac{3}{2\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} \\ &= \frac{3\sqrt{3}}{6}\end{aligned}$$

$$\text{i.e. } \cos \angle BAO = \frac{\sqrt{3}}{2}$$

$$\therefore \angle BAO = \frac{\pi}{6}.$$

$$\text{(ii)} \quad A = \frac{1}{2} r^2 \theta$$

$$r = r\sqrt{3}$$

$$\theta = \angle BAC = \frac{\pi}{6} + \frac{\pi}{6} = \frac{2\pi}{6} = \frac{\pi}{3}$$

$$(\angle BAC = \angle BAO + \angle CAO)$$

$$\text{and } \angle CAO = \frac{\pi}{6} \text{ as } \triangle BAO \equiv \triangle CAO$$

$$\therefore A = \frac{1}{2} (r\sqrt{3})^2 \left(\frac{\pi}{3}\right)$$

$$= \frac{1}{2} \cdot 3r^2 \cdot \frac{\pi}{3}$$

$$= \frac{\pi r^2}{2}.$$

$$\text{(iii)} \quad A = \frac{1}{2} r^2 \theta$$

$$r = r$$

$$\theta = \angle BOC = 120^\circ = \frac{2\pi}{3}$$

(angle at centre is twice
angle at circumference)

$$\therefore A = \frac{1}{2} r^2 \left(\frac{2\pi}{3}\right)$$

$$= \frac{\pi r^2}{3}$$

(iv) Area of shaded region
= Area of Sector BOC - (Area of Sector BAC - Area $\triangle$'s AOB and AOC)

$$= \frac{\pi r^2}{3} - \left[\frac{\pi r^2}{2} - 2 \left(\frac{\sqrt{3} r^2}{4} \right) \right]$$

$$= \frac{\pi r^2}{3} - \left(\frac{\pi r^2}{2} - \frac{\sqrt{3} r^2}{2} \right)$$

$$= \frac{\pi r^2}{3} - \frac{\pi r^2}{2} + \frac{\sqrt{3} r^2}{2}$$

$$= r^2 \left(\frac{\pi}{3} - \frac{\pi}{2} + \frac{\sqrt{3}}{2} \right)$$

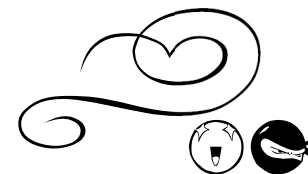
$$= r^2 \left(\frac{\sqrt{3}}{2} - \frac{\pi}{6} \right) \text{ units}^2.$$





Trigonometry

BOS

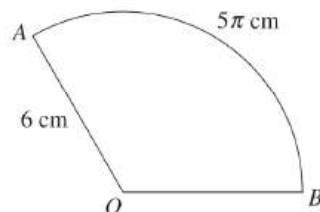


2004

Q11

AOB is a sector of a circle, centre O and radius 6 cm.
The length of the arc AB is 5π cm.

Calculate the exact area of the sector AOB .



$$\ell = r\theta \quad (\theta \text{ in radians})$$

$$5\pi = 6\theta$$

$$\theta = \frac{5\pi}{6}$$

$$A = \frac{1}{2}r^2\theta \quad (\theta \text{ in radians})$$

$$= \frac{1}{2} \times 6^2 \times \frac{5\pi}{6}$$

$$= \frac{1}{2} \times 36 \times \frac{5\pi}{6}$$

$$= 15\pi.$$

$$\therefore \text{Area is } 15\pi \text{ cm}^2.$$

2004

Q12

- (i) Show that $\cos\theta \tan\theta = \sin\theta$.
(ii) Hence solve $8\sin\theta \cos\theta \tan\theta = \text{cosec}\theta$ for $0 \leq \theta \leq 2\pi$.

(i) $\cos\theta \tan\theta = \sin\theta$.

$$\begin{aligned} \text{LHS} &= \cos\theta \tan\theta \\ &= \cos\theta \cdot \frac{\sin\theta}{\cos\theta} \\ &= \sin\theta \\ &= \text{RHS}. \end{aligned}$$

(ii) $8\sin\theta \cos\theta \tan\theta = \text{cosec}\theta$

$$8\sin\theta \times \sin\theta = \text{cosec}\theta$$

$$8\sin^2\theta = \text{cosec}\theta$$

$$8\sin^2\theta = \frac{1}{\sin\theta}$$

$$8\sin^3\theta = 1$$

$$\sin^3\theta = \frac{1}{8}$$

$$\sin\theta = \sqrt[3]{\frac{1}{8}}$$

$$\sin\theta = \frac{1}{2}$$

[The related angle is $\frac{\pi}{6}$]

In the 1st quadrant, $\theta = \frac{\pi}{6}$.

In the 2nd quadrant, $\theta = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$.

$\therefore \theta = \frac{\pi}{6}$ or $\frac{5\pi}{6}$.

2

2005

Q13

Solve $\cos\theta = \frac{1}{\sqrt{2}}$ for $0 \leq \theta \leq 2\pi$.

$$\cos\theta = \frac{1}{\sqrt{2}} \quad (0 \leq \theta \leq 2\pi).$$

θ lies in 1st, 4th quadrants: $\theta = \frac{\pi}{4}, 2\pi - \frac{\pi}{4}$

$$\theta = \frac{\pi}{4}, \frac{7\pi}{4}.$$

2

2005

Q14

The lengths of the sides of a triangle are 7 cm, 8 cm and 13 cm.

- (i) Find the size of the angle opposite the longest side.
(ii) Find the area of the triangle.

2

1

- (i) The largest angle θ will be opposite the longest side, 13 cm.

Using the Cosine Rule:

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$\cos\theta = \frac{7^2 + 8^2 - 13^2}{2 \times 7 \times 8}$$

$$= -0.5.$$

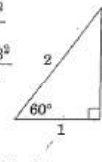
$$\text{Acute } \theta = \cos^{-1}\left(\frac{1}{2}\right)$$

$$= 60^\circ.$$

Since $\cos\theta$ is negative, θ is obtuse.

$$\theta = 180^\circ - 60^\circ$$

$$\theta = 120^\circ.$$

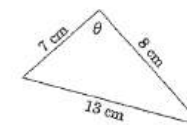


- (ii) Area = $\frac{1}{2}ab \sin C$

$$= \frac{1}{2} \times 7 \times 8 \times \sin 120^\circ$$

$$= \frac{1}{2} \times 7 \times 8 \times \frac{\sqrt{3}}{2}$$

$$= 14\sqrt{3} \text{ cm}^2.$$

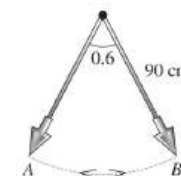


2005

Q15

A pendulum is 90 cm long and swings through an angle of 0.6 radians. The extreme positions of the pendulum are indicated by the points A and B in the diagram.

- (i) Find the length of the arc AB .
(ii) Find the straight-line distance between the extreme positions of the pendulum.
(iii) Find the area of the sector swept out by the pendulum.



1

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(i) Arc $AB(\ell) = r\theta$
 $= 90 \times 0.6$
 $= 54 \text{ cm}.$

(ii) By the cosine rule,
 $a^2 = b^2 + c^2 - 2bc \cos A$.
In this case $b = 90$, $c = 90$, $A = 0.6$ radians.
 $AB^2 = 90^2 + 90^2 - 2 \times 90 \times 90 \times \cos 0.6$
 $= 2829.563 \dots$

Distance $AB = 53.2 \text{ cm}$ (1 dec. place).

Note: The calculator must be set to radians to enter 0.6 radians. Alternatively, to enter the angle in degrees,

$$0.6 \text{ radians} = 0.6 \times \frac{180^\circ}{\pi}.$$

(iii) Area sector = $\frac{1}{2}r^2\theta$.

In this case $r = 90$, $\theta = 0.6$ radians.

$$\text{Area sector} = \frac{1}{2} \times 90^2 \times 0.6$$

$$= 2430.$$

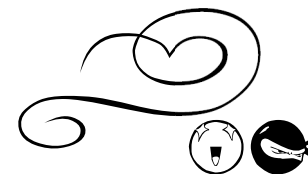
$\therefore$ Area swept out by pendulum is 2430 cm^2 .





Trigonometry

BOS



NAME: _____

2006

Q16

Find the value of θ in the diagram. Give your answer to the nearest degree.



$$\begin{aligned}\frac{\sin A}{a} &= \frac{\sin B}{b} \\ \frac{\sin \theta}{5} &= \frac{\sin 33^\circ}{9} \\ \sin \theta &= \frac{5 \sin 33^\circ}{9} \\ \theta &= 17.6 \dots^\circ \\ &= 18^\circ \text{ (to the nearest degree)}.\end{aligned}$$

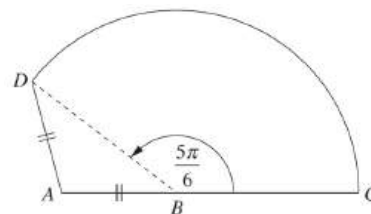
2

2006

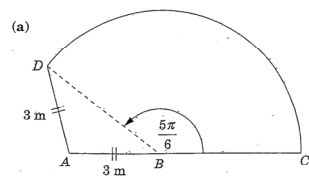
Q17

In the diagram, $ABCD$ represents a garden. The sector BCD has centre B and $\angle DBC = \frac{5\pi}{6}$. The points A , B and C lie on a straight line and $AB = AD = 3$ metres.

- Show that $\angle DAB = \frac{2\pi}{3}$.
- Find the length of BD .
- Find the area of the garden $ABCD$.



**1
2
2**



$$\begin{aligned}\text{(i)} \quad \angle DBA &= \pi - \frac{5\pi}{6} \quad (\angle ABC \text{ is a straight line}) \\ &= \frac{\pi}{6} \\ \angle BDA &= \frac{\pi}{6} \quad (\text{equal } \angle \text{s opp. equal sides of isosceles } \triangle BDA) \\ \therefore \angle DAB &= \pi - \left(\frac{\pi}{6} + \frac{\pi}{6}\right) \quad (\angle \text{sum of } \triangle BDA \text{ is } \pi) \\ &= \pi - \frac{\pi}{3} \\ \angle DAB &= \frac{2\pi}{3}\end{aligned}$$

$$\begin{aligned}\text{(ii) Using the cosine rule} \\ a^2 &= b^2 + c^2 - 2bc \cos A \\ BD^2 &= 3^2 + 3^2 - 2 \times 3 \times 3 \times \cos \frac{2\pi}{3} \\ &= 9 + 9 - 18 \times \left(-\frac{1}{2}\right) \\ &= 18 + 9 \\ BD^2 &= 27 \\ BD &= \sqrt{27} \text{ m or } 3\sqrt{3} \text{ m or } 5.196 \dots \text{ m}.\end{aligned}$$

$$\begin{aligned}\text{(iii) Using } A &= \frac{1}{2} ab \sin C, \\ \text{area } \triangle ABD &= \frac{1}{2} \times 3 \times 3 \times \sin \frac{2\pi}{3} \\ &= \frac{1}{2} \times 3 \times 3 \times \frac{\sqrt{3}}{2} \\ &= \frac{9\sqrt{3}}{4} \\ \text{Using } A &= \frac{1}{2} r^2 \theta, r = \sqrt{27}, \\ \text{area of sector } BDC &= \frac{1}{2} \times (\sqrt{27})^2 \times \frac{5\pi}{6} \\ &= \frac{1}{2} \times 27 \times \frac{5\pi}{6} \\ &= \frac{45\pi}{4} \\ \text{Garden area} &= \frac{9\sqrt{3}}{4} + \frac{45\pi}{4} \text{ m}^2 \\ &= \frac{9\sqrt{3} + 45\pi}{4} \text{ m}^2 \\ &= 39.240 \dots \text{ m}^2.\end{aligned}$$

2007

Q18

Solve $\sqrt{2} \sin x = 1$ for $0 \leq x \leq 2\pi$.

$$|\sqrt{2} \sin x| = 1$$

$$\sin x = \frac{1}{\sqrt{2}} \quad x = \frac{\pi}{4}, \frac{3\pi}{4}$$

2

2007

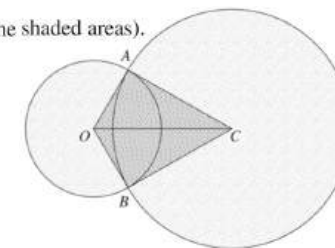
Q19

An advertising logo is formed from two circles, which intersect as shown. The circles intersect at A and B and have centres at O and C .

The radius of the circle centred at O is 1 metre and the radius of the circle centred at C is $\sqrt{3}$ metres. The length of OC is 2 metres.

- Use Pythagoras' theorem to show that $\angle OAC = \frac{\pi}{2}$.
- Find $\angle ACO$ and $\angle AOC$.
- Find the area of the quadrilateral $AOBC$.
- Find the area of the major sector ACB .
- Find the total area of the logo (the sum of all the shaded areas).

**1
2
1
1
2**



$$\text{(c)(i) } OC^2 = OA^2 + AC^2$$

$$2^2 = 1^2 + (\sqrt{3})^2$$

$$4 = 4 \therefore \angle AOC = \frac{\pi}{2}$$

$$\begin{aligned}\text{(iv) } (A_{\text{sector } ACB}) &= \frac{1}{2} r^2 \theta \\ &= \frac{1}{2} (\sqrt{3})^2 \frac{\pi}{3} \\ &= \frac{\pi}{2} u^2\end{aligned}$$

$$\text{(ii) } \tan \angle ACO = \frac{1}{\sqrt{3}}$$

$$\therefore \angle ACO = \frac{\pi}{6}$$

$$\tan \angle AOC = \frac{\sqrt{3}}{1} \therefore \angle AOC = \frac{\pi}{3}$$

$$\begin{aligned}\text{(v) Area} &= \text{Area of larger circle} - A_{\text{sector } ACB} \\ &\quad + \text{Area of smaller circle} - A_{\text{sector } AOB} + A_{\text{AOBC}} \\ &= \pi \cdot 3 - \frac{\pi}{2} + \pi \cdot 1 + \frac{1}{2} \cdot 1^2 \cdot \frac{2\pi}{3} + \sqrt{3} \\ &= \frac{5\pi}{2} + \frac{2\pi}{3} + \sqrt{3} \\ &= \frac{19\pi}{6} + \sqrt{3}\end{aligned}$$

$$\text{(iii) } A = 2 \cdot \frac{1}{2} \cdot 1 \cdot \sqrt{3} = \sqrt{3} u^2$$

2008

Evaluate $2 \cos \frac{\pi}{5}$ correct to three significant figures.

1.62

1

Q20

2008

Q21

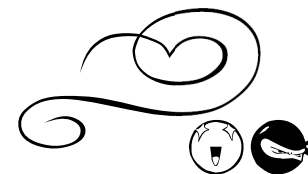
Solve $2 \sin^2 \frac{x}{3} = 1$ for $-\pi \leq x \leq \pi$.

$$2 \sin^2 \frac{x}{3} = 1 \text{ and } -\pi \leq x \leq \pi$$

$$\begin{aligned}1 - 2 \sin^2 \frac{x}{3} &= 0, \therefore \cos \frac{2x}{3} = 0 \text{ and } -\frac{2\pi}{3} \leq \frac{2x}{3} \leq \frac{2\pi}{3} \\ \therefore \frac{2x}{3} &= \pm \frac{\pi}{2}, x = \pm \frac{3\pi}{4}.\end{aligned}$$

1





2008

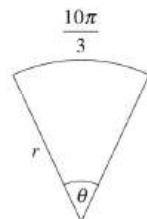
The diagram shows a sector with radius r and angle θ where $0 < \theta \leq 2\pi$.

The arc length is $\frac{10\pi}{3}$.

Q22

(i) Show that $r \geq \frac{5}{3}$.

(ii) Calculate the area of the sector when $r = 4$.



Q7bi $r\theta = \frac{10\pi}{3}$, $\therefore \theta = \frac{10\pi}{3r}$.

Since $0 < \theta \leq 2\pi$, $\therefore 0 < \frac{10\pi}{3r} \leq 2\pi$, $0 < \frac{5}{3r} \leq 1$,

$0 < 5 \leq 3r$, $\therefore r \geq \frac{5}{3}$.

Q7bii Area = $\frac{1}{2}r^2\theta = \frac{1}{2}r(r\theta) = \frac{1}{2}(4)\left(\frac{10\pi}{3}\right) = \frac{20\pi}{3}$ square units

2009

Q23

Find the exact value of θ such that $2\cos\theta = 1$, where $0 \leq \theta \leq \frac{\pi}{2}$.

$2\cos\theta = 1$, $\theta = \cos^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{3}$.

2009

Q24

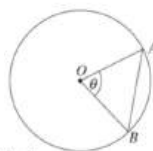
The diagram shows a circle with centre O and radius 2 centimetres. The points A and B lie on the circumference of the circle and $\angle AOB = \theta$.

(i) There are two possible values of θ for which the area of $\triangle AOB$ is $\sqrt{3}$ square centimetres. One value is $\frac{\pi}{3}$. Find the other value.

(ii) Suppose that $\theta = \frac{\pi}{3}$.

(1) Find the area of the sector AOB .

(2) Find the exact length of the perimeter of the minor segment bounded by the chord AB and the arc AB .



Q5ci $A = \frac{1}{2}r^2\sin\theta$, where $0 \leq \theta \leq \pi$.

Q5cii(1) Area of sector $AOB = \frac{\theta}{2\pi} \times \pi r^2 = \frac{2\pi}{3} \text{ cm}^2$.

$\sqrt{3} = \frac{1}{2} \times 2^2 \sin\theta$, $\sin\theta = \frac{\sqrt{3}}{2}$, $\theta = \frac{\pi}{3}$ or $\frac{2\pi}{3}$.

Q5cii(2) Chord $AB = \sqrt{2^2 + 2^2 - 2 \times 2 \times 2 \cos\frac{\pi}{3}} = 2 \text{ cm}$.

The other value of θ is $\frac{2\pi}{3}$.

Arc $AB = r\theta = 2 \times \frac{\pi}{3} = \frac{2\pi}{3} \text{ cm}$.

Perimeter of minor segment = $2 + \frac{2\pi}{3} \text{ cm}$.

2010

Q25

(i) Prove that $\sec^2 x + \sec x \tan x = \frac{1 + \sin x}{\cos^2 x}$.

2

(ii) Hence prove that $\sec^2 x + \sec x \tan x = \frac{1}{1 - \sin x}$.

2

Q5bi $RHS = \frac{1 + \sin x}{\cos^2 x} = \frac{1}{\cos^2 x} + \frac{\sin x}{\cos^2 x}$

$= \left(\frac{1}{\cos x}\right)^2 + \left(\frac{1}{\cos x}\right)\left(\frac{\sin x}{\cos x}\right) = \sec^2 x + \sec x \tan x = LHS$

Q5bii $\sec^2 x + \sec x \tan x = \frac{1 + \sin x}{\cos^2 x} = \frac{1 + \sin x}{1 - \sin^2 x} = \frac{1 + \sin x}{(1 - \sin x)(1 + \sin x)} = \frac{1}{1 - \sin x}$

2010

Q26

The diagram shows a circle with centre O and radius 5 cm.

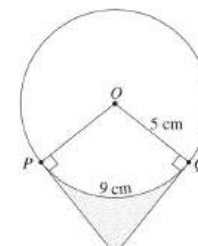
The length of the arc PQ is 9 cm. Lines drawn perpendicular to OP and OQ at P and Q respectively meet at T .

(i) Find $\angle POQ$ in radians.

(ii) Prove that $\triangle OPT$ is congruent to $\triangle OQT$.

(iii) Find the length of PT .

(iv) Find the area of the shaded region.



1

2

Q6bi $9 = 5\angle POQ$, $\therefore \angle POQ = \frac{9}{5} = 1.8$

Q6biii $\angle POT = \frac{1.8}{2} = 0.9$, length of $PT = 5 \tan(0.9) \approx 6.30 \text{ cm}$

Q6bii $OP = OQ = \text{radius}$

OT is a common side for $\triangle OPT$ and $\triangle OQT$

$\angle OPT = \angle OQT = 90^\circ$, $\therefore PT = QT$ (Pythagoras)

$\therefore \triangle OPT$ and $\triangle OQT$ are congruent because they have equal corresponding sides.

Q6biv Area of the shaded region

$= 5 \times 6.3 - \frac{1.8}{2} \times \pi \times 5^2 \approx 9.00 \text{ cm}^2$

2011

Q27

Find the exact values of x such that $2\sin x = -\sqrt{3}$, where $0 \leq x \leq 2\pi$.

2

Q2b $2\sin x = -\sqrt{3}$ and $0 \leq x \leq 2\pi$, $\sin x = -\frac{\sqrt{3}}{2}$

$x = \frac{4\pi}{3}, \frac{5\pi}{3}$

2011

Q28

In the diagram, the shop at S is 20 kilometres across the bay from the post office at P . The distance from the shop to the lighthouse at L is 22 kilometres and $\angle SPL$ is 60° .

Let the distance PL be x kilometres.

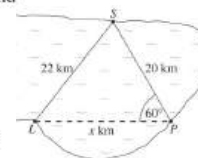
(i) Use the cosine rule to show that $x^2 - 20x - 84 = 0$.

(ii) Hence, find the distance from the post office to the lighthouse. Give your answer correct to the nearest kilometre.

Q8ai $x^2 + 20^2 - 2(x)(20)\cos 60^\circ = 22^2$
 $x^2 + 400 - 20x = 484$, $\therefore x^2 - 20x - 84 = 0$

Q8aii $x = \frac{-(-20) \pm \sqrt{(-20)^2 - 4(1)(-84)}}{2} = \frac{20 \pm \sqrt{736}}{2}$

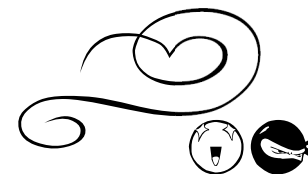
$\therefore PL = \frac{20 + \sqrt{736}}{2} \approx 24 \text{ km}$



1

2





2012

Q29

What are the solutions of $\sqrt{3}\tan x = -1$ for $0 \leq x \leq 2\pi$?

(A) $\frac{2\pi}{3}$ and $\frac{4\pi}{3}$

(C) $\frac{5\pi}{6}$ and $\frac{7\pi}{6}$

(B) $\frac{2\pi}{3}$ and $\frac{5\pi}{3}$

(D) $\frac{5\pi}{6}$ and $\frac{11\pi}{6}$

Q6 $\sqrt{3}\tan x = -1, 0 \leq x \leq 2\pi$
 $\tan x = -\frac{1}{\sqrt{3}}, x = \frac{5\pi}{6}, \frac{11\pi}{6}$

1

2012

Q30

The area of a sector of a circle of radius 6 cm is 50 cm^2 . Find the length of the arc of the sector.

Q11f $A = \frac{1}{2}r^2\theta, \therefore A = \frac{1}{2}r\ell$ where $\ell = r\theta$ is the length of the arc,
 $\therefore 50 = \frac{1}{2} \times 6 \times \ell, \ell = \frac{50}{3} \text{ cm}$

1

2013

Q31

The region ABC is a sector of a circle with radius 30 cm, centred at C . The angle of the sector is θ . The arc DE lies on a circle also centred at C , as shown in the diagram.

The arc DE divides the sector ABC into two regions of equal area. Find the exact length of the interval CD .

2

Area sector $CDE \Rightarrow A_1 = \frac{1}{2} \cdot CD^2 \cdot \theta$

Area sector $CAB \Rightarrow A_2 = \frac{1}{2} \cdot 30^2 \cdot \theta = 450\theta$

Given: $A_1 = \text{Area } ADBE = (A_2 - A_1)$

So $A_1 = A_2 - A_1 \Rightarrow 2A_1 = A_2 \Rightarrow \frac{1}{2}CD^2 \cdot \theta = 450\theta$
 $CD^2 = 450 \Rightarrow CD = 15\sqrt{2} \text{ cm}$

2013

Q32

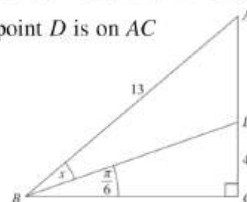
The right-angled triangle ABC has hypotenuse $AB = 13$. The point D is on AC such that $DC = 4$, $\angle DBC = \frac{\pi}{6}$ and $\angle ABD = x$.

Using the sine rule, or otherwise, find the exact value of $\sin x$.

Q14c $\overline{BC} = \frac{4}{\tan \frac{\pi}{6}} = 4\sqrt{3}, \overline{AC} = \sqrt{13^2 - (4\sqrt{3})^2} = 11$

$\therefore \overline{AD} = 11 - 4 = 7, \overline{BD} = \frac{4}{\sin \frac{\pi}{6}} = 8, \angle ADB = \frac{\pi}{2} + \frac{\pi}{6} = \frac{2\pi}{3}$

The sine rule: $\frac{\sin x}{7} = \frac{\sin \frac{2\pi}{3}}{13}, \therefore \sin x = \frac{7\sqrt{3}}{26}$



2

2014

Q33

How many solutions of the equation $(\sin x - 1)(\tan x + 2) = 0$ lie between 0 and 2π ?

(A) 1

(B) 2

(C) 3

(D) 4

1

Q7 $(\sin x - 1)(\tan x + 2) = 0, \therefore \tan x + 2 = 0$ which has 2 solutions between 0 and 2π . $\sin x - 1 = 0$ has 1 solution, i.e. $x = \frac{\pi}{2}$ which is undefined for $\tan x$.
 $\therefore$ Total number of solutions is 2.

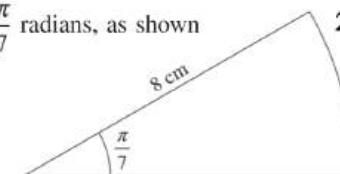
2014

Q34

The angle of a sector in a circle of radius 8 cm is $\frac{\pi}{7}$ radians, as shown. Find the exact value of the perimeter of the sector.

Q11g Arc length $= r\theta = 8 \times \frac{\pi}{7} = \frac{8\pi}{7} \text{ cm}$

Perimeter $= 8 + 8 + \frac{8\pi}{7} = 8\left(2 + \frac{\pi}{7}\right) \text{ cm}$



2

2014

Q35

Find all solutions of $2\sin^2 x + \cos x - 2 = 0$, where $0 \leq x \leq 2\pi$.

Q15a $2\sin^2 x + \cos x - 2 = 0, 0 \leq x \leq 2\pi$

$2(1 - \cos^2 x) + \cos x - 2 = 0$

$\cos x - 2\cos^2 x = 0, \cos x(1 - 2\cos x) = 0$

$\therefore \cos x = 0, x = \frac{\pi}{2}, \frac{3\pi}{2}$ or $1 - 2\cos x = 0, x = \frac{\pi}{3}, \frac{5\pi}{3}$

The solutions are $x = \frac{\pi}{3}, \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{3}$

2

2014

Q36

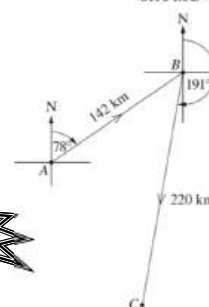
Chris leaves island A in a boat and sails 142 km on a bearing of 078° to island B. Chris then sails on a bearing of 191° for 220 km to island C, as shown.

(i) Show that the distance from island C to island A is approximately 210 km.

(ii) Chris wants to sail from island C directly to island A. On what bearing should Chris sail? Give your answer correct to the nearest degree.

2

3



Q13di

$\theta = 78 - 11 = 67^\circ$

$CA = \sqrt{142^2 + 220^2 - 2(142)(220)\cos 67^\circ}$

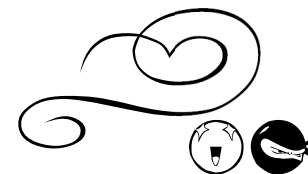
≈ 210.12

$\therefore CA$ is approximately 210 km

Q13dii $\frac{\sin \phi}{142} = \frac{\sin 67^\circ}{210.12}, \therefore \phi = 38.47^\circ$

$\therefore$ bearing is $360^\circ - (38.47^\circ - 11^\circ) = 333^\circ$





2015

Find the solutions of $2\sin\theta = 1$ for $0 \leq \theta \leq 2\pi$. $2\sin\theta = 1$

Q37

$$\Rightarrow \theta = \frac{\pi}{6}, \frac{5\pi}{6} \quad (0 \leq \theta \leq 2\pi)$$

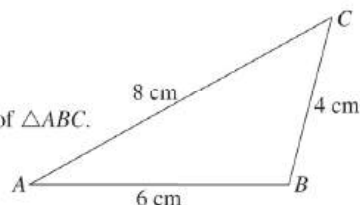
2015

The diagram shows $\triangle ABC$ with sides $AB = 6$ cm, $BC = 4$ cm and $AC = 8$ cm.

Q38

(i) Show that $\cos A = \frac{7}{8}$.

(ii) By finding the exact value of $\sin A$, determine the exact value of the area of $\triangle ABC$.



Question 13 (a) (i)

By cosine rule, $\cos A = \frac{6^2 + 8^2 - 4^2}{2(6)(8)} = \frac{7}{8}$

Question 13 (a) (ii)

$\cos A = \frac{7}{8} \Rightarrow \sin A = \frac{\sqrt{15}}{8}$ (by constructing a right-angled triangle)

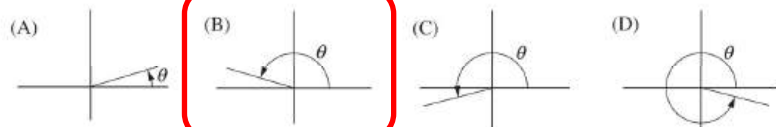
So area = $\frac{1}{2}bc \sin A = \frac{24 \times \sqrt{15}}{8} = 3\sqrt{15}$.

2016

For the angle θ , $\sin\theta = \frac{7}{25}$ and $\cos\theta = -\frac{24}{25}$.

Q39

Which diagram best shows the angle θ ?



2016

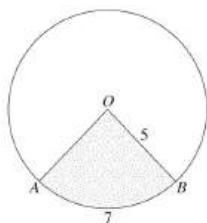
The circle centred at O has radius 5. Arc AB has length 7 as shown in the diagram.

Q40

What is the area of the shaded sector OAB ?

- (A) $\frac{35}{2}$ (C) $\frac{125}{14}$
(B) $\frac{35}{2}\pi$ (D) $\frac{125}{14}\pi$

Area of shaded region = $\frac{7}{2\pi \times 5} \times \pi \times 5^2 = \frac{35}{2}$



2

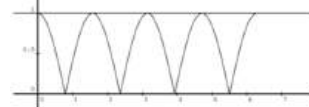
2016

How many solutions does the equation $|\cos(2x)| = 1$ have for $0 \leq x \leq 2\pi$?

1

Q41

- (A) 1 (B) 3 (C) 4 (D) 5



2

2017

Which expression is equivalent to $\tan\theta + \cot\theta$?

1

Q42

- A. $\operatorname{cosec}\theta + \sec\theta$ B. $\sec\theta \operatorname{cosec}\theta$ C. 2 D. 1

2017

In the diagram, OAB is a sector of the circle with centre O and radius 6 cm, where $\angle AOB = 30^\circ$.

Q43

- (i) Find the exact value of the area of the triangle OAB .
(ii) Find the exact value of the area of the shaded segment.

Question 11 (e) (i)

Sample answer:

$$\begin{aligned} A &= \frac{1}{2}ab \sin C \\ &= \frac{1}{2} \times 6 \times 6 \times \sin 30^\circ \\ &= \frac{1}{2} \times 36 \times \frac{1}{2} \end{aligned}$$

$\therefore \text{Area} = 9 \text{ cm}^2$

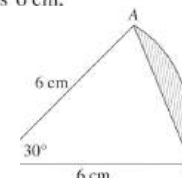
Question 11 (e) (ii)

Sample answer:

Shaded area = Area of sector OAB - Area of $\triangle OAB$

$$\begin{aligned} &= \frac{\pi}{12} (6^2) - 9 \\ &= \frac{36\pi}{12} - 9 \end{aligned}$$

$\therefore \text{Area} = (3\pi - 9) \text{ cm}^2$



1
1

1

2017

Using the cosine rule, find the value of x in the following diagram.

Q44

Using cosine rule:

$$13^2 = (x+4)^2 + (x-4)^2 - 2(x+4)(x-4)\cos 60^\circ$$

$$169 = x^2 + 8x + 16 + x^2 - 8x + 16 - 2(x^2 - 16) \times \frac{1}{2}$$

$$= 2x^2 + 32 - x^2 + 16$$

$$= x^2 + 48$$

$$121 = x^2$$

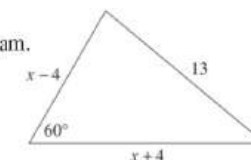
$$\therefore x = \pm\sqrt{121}$$

$$= \pm 11$$

But $x - 4 > 0$ since it is a length

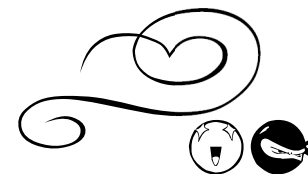
So $x > 4$

$$\therefore x = 11$$



1
1





2016

Q45

Square tiles of side length 20 cm are being used to tile a bathroom.

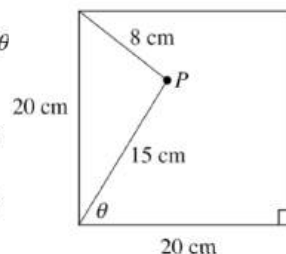
The tiler needs to drill a hole in one of the tiles at a point P which is 8 cm from one corner and 15 cm from an adjacent corner.

To locate the point P the tiler needs to know the size of the angle θ

Find the size of the angle θ to the nearest degree.

Q12c The cosine rule: $20^2 + 15^2 - 2 \times 20 \times 15 \cos(90^\circ - \theta) = 8^2$

$$\cos(90^\circ - \theta) = \frac{20^2 + 15^2 - 8^2}{600} = 0.935, \quad 90^\circ - \theta \approx 21^\circ, \quad \theta \approx 69$$



2

2018

Q46

A ship travels from Port A on a bearing of 050° for 320 km to Port B. It then travels on a bearing of 120° for 190 km to Port C.

(i) What is the size of $\angle ABC$?

(ii) What is the distance from Port A to Port C?

Answer to the nearest 10 kilometres.

Sample answer:
 $\angle ABC = 50^\circ + 60^\circ$

(a) (i) $= 110^\circ$

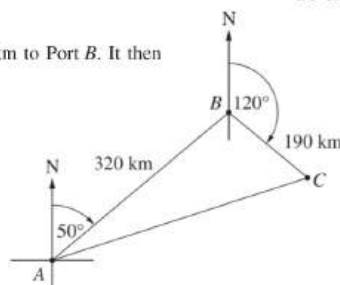
(a) (ii) Sample answer:

$$d^2 = 320^2 + 190^2 - 2(320)(190)\cos 110^\circ$$

$$d = \sqrt{320^2 + 190^2 - 2(320)(190)\cos 110^\circ}$$

$$d = 424.3697084$$

$$d = 420 \text{ km}$$



1

2

2018

Q47

In $\triangle KLM$, KL has length 3, LM has length 6 and $\angle KLM$ is 60° . The point N is chosen on side KM so that LN bisects $\angle KLM$. The length LN is x .

(i) Find the exact value of the area of $\triangle KLM$.

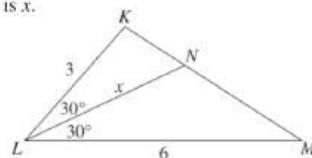
(ii) Hence, or otherwise, find the exact value of x .

$$\frac{1}{2} \times 3 \times x \times \sin 30^\circ + \frac{1}{2} \times 6 \times x \times \sin 30^\circ = \frac{9\sqrt{3}}{2}$$

$$\frac{3x}{2} + 3x = 9\sqrt{3}$$

$$\frac{9x}{2} = 9\sqrt{3}$$

$$x = 2\sqrt{3}$$



1

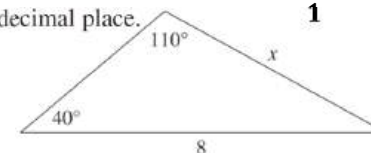
2

2019

Q49

Using the sine rule, find the value of x correct to one decimal place.

$$\frac{x}{\sin 40^\circ} = \frac{8}{\sin 110^\circ}, \quad x = 5.5$$



1

2019

Q50

Solve $2 \sin x \cos x = \sin x$ for $0 \leq x \leq 2\pi$.

Q13a $2 \sin x \cos x - \sin x = 0$, $\sin x(2 \cos x - 1) = 0$ and $0 \leq x \leq 2\pi$

$$\therefore \sin x = 0 \text{ or } \cos x = \frac{1}{2} \therefore x = 0, \frac{\pi}{3}, \pi, \frac{5\pi}{3}, 2\pi$$

3

2019

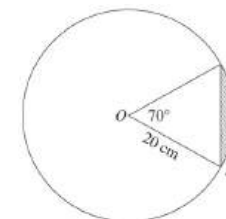
Q51

The diagram shows a circle with centre O and radius 20 cm. The points A and B lie on the circle such that $\angle AOB = 70^\circ$.

Find the perimeter of the shaded segment, giving your answer correct to one decimal place.

Q13b Perimeter = arc AB + chord AB

$$= \frac{70}{360} \times 2\pi \times 20 + 2 \times 20 \sin 35^\circ \approx 47.4 \text{ cm}$$



3

2019

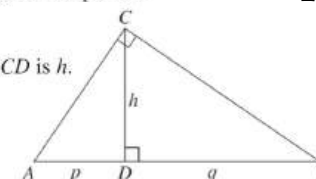
Q52

The triangle ABC is a right-angled triangle with the right angle at C . The point D is chosen on AB so that CD is perpendicular to AB .

The length of AD is p , the length of BD is q and the length of CD is h . Show that $h = \sqrt{pq}$.

$$\text{Q15b } \overline{AC}^2 + \overline{BC}^2 = \overline{AB}^2, \therefore p^2 + h^2 + h^2 + q^2 = (p+q)^2$$

$$\therefore h^2 = pq, \quad h = \sqrt{pq}$$



2

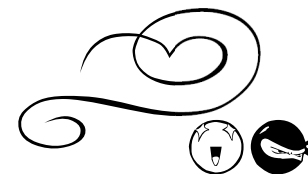
2020

Q53

Which interval gives the range of the function $y = 5 + 2 \cos 3x$?

- A. $[2, 8]$ B. $[3, 7]$ C. $[4, 6]$ D. $[5, 9]$

2

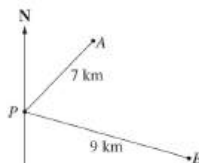


2020

Q54

Mr Ali, Ms Brown and a group of students were camping at the site located at P . Mr Ali walked with some of the students on a bearing of 035° for 7 km to location A . Ms Brown, with the rest of the students, walked on a bearing of 100° for 9 km to location B .

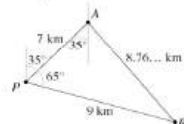
- Show that the angle APB is 65° .
- Find the distance AB .
- Find the bearing of Ms Brown's group from Mr Ali's group. Give your answer correct to the nearest degree.



1
2
2

Question 15 (a) Question 15 (b)

Sample answer:
 $100^\circ - 35^\circ = 65^\circ$



$$AB^2 = 7^2 + 9^2 - 2 \times 7 \times 9 \cos 65^\circ$$

$$= 76.75...$$

$$AB = 8.76 \text{ km}$$

Question 15 (c)

Sample answer:

$$\frac{\sin A}{9} = \frac{\sin 65^\circ}{8.76...}$$

$$A = 68.6...$$

$$\approx 69^\circ$$

$$\text{Bearing} = 180^\circ - (69^\circ - 35^\circ)$$

$$= 146^\circ$$

2020

Q55

Prove that $\sec \theta - \cos \theta = \sin \theta \tan \theta$.

$$\sec \theta - \cos \theta = \sin \theta \tan \theta$$

$$\text{LHS} = \frac{1}{\cos \theta} - \cos \theta$$

$$= \frac{1 - \cos^2 \theta}{\cos \theta} = \sin \theta \times \frac{\sin \theta}{\cos \theta}$$

$$= \frac{\sin^2 \theta}{\cos \theta} = \sin \theta \times \tan \theta$$

$$= \text{RHS}$$

2

2020

Q56

The diagram shows a regular decagon (ten-sided shape with all sides equal and all interior angles equal). The decagon has centre O .

The perimeter of the shape is 80 cm.

By considering triangle OAB , calculate the area of the ten-sided shape. Give your answer in square centimetres correct to one decimal place.

Sample answer:

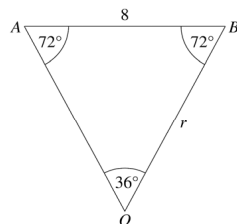
$$AB = 80 \div 10$$

$$= 8 \text{ cm}$$

$$\angle AOB = 360^\circ \div 10 = 36^\circ$$

$$\angle OAB = \frac{180^\circ - 36^\circ}{2}$$

$$= 72^\circ$$

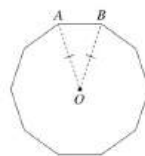


$$\frac{r}{\sin 72^\circ} = \frac{8}{\sin 36^\circ}$$

$$r = 12.944...$$

$$\text{Area} = 10 \times \frac{1}{2} \times (12.944...) \times \sin 36^\circ$$

$$\approx 492.4 \text{ cm}^2$$



4